

Sample Calculations : Heat Transfer in a Stirred Tank

The example calculations will be carried out for a trial at 615 RPM, no baffles
Tank temperature = 66°C. See data attached.

Calculation Of Q_{13}

The equation for Q_{13} is $Q_{13} = Q_{coil} + Q_{hx}$ where:

$$\begin{aligned}Q_{coil} &= m_{coil} C_p (t_{4,out} - t_{4,in}) \\Q_{hx} &= m_{ves} C_p (t_3 - t_{hx,out})\end{aligned}$$

Where m_{coil} is the mass flow rate of the cooling water in the coils, C_p is the heat capacity of water, $t_{4,in}$ and $t_{4,out}$ are the inlet and outlet water temperatures in the cooling coils, m_{ves} is the mass flow rate of the vessel recirculation water in the heat exchanger, t_3 is the vessel water temperature that should be equal to the inlet recirculation water temperature to the heat exchanger, and $t_{hx,out}$ is the outlet recirculation water temperature from the heat exchanger. Heat loss to the surroundings, Q_{surr} , is assumed to be negligible.

Taking the first trial at 615 RPM a calculation will be performed for Q_{hx}

$$Q_{hx} = .050 * 4184 * (338 - 328) = 2092 \text{ W}$$

Converting this number to kW the answer is 2.092 kW

Then a similar equation will be used for Q_{coils} where m is the cooling water flow, C_p is the heat capacity of water and the change in temperature is the change in temperature of cooling water.

$$Q_{coils} = .047 * 4184 * (319.11 - 294.11) = 4916.2 \text{ W}$$

Then converting to kW the answer is 4.916 kW

Then using the equation $Q_{13} = Q_{coils} + Q_{hx}$

$$Q_{13} = 2.092 + 4.916 = 7.008 \text{ kW}$$

Calculation of Reynolds number

The following equation was used for impeller Reynolds number

$$\frac{D_i^2 n \rho}{\mu} \equiv Re_{imp} = \text{impeller Reynolds number}$$

Taking the measurements for the diameter, and converting the impeller speed to rotations per second the Re_{imp}

$$Re_{imp} = \frac{980 * \left(\frac{615}{60}\right) * (.1012^2)}{0.000432} = 238137.19$$

Calculation of Prandtl Number

$$\frac{C_p \mu}{k} \equiv Pr = \text{fluid Prandtl number}$$

The calculation of the fluid Prandtl number is as follows the heat capacity of water, the viscosity of water and finally, the thermal conductivity of the coil. The fluid Prandtl number is

$$Pr = \frac{4184 * 0.000432}{0.645} = 2.80$$

Calculation of Nusselt Number

$$Nu_T = .36 * Re_{imp}^{\frac{2}{3}} * Pr^{\frac{1}{3}}$$

Using the results from the previous calculations the Nusselt Number is as follows:

$$Nu_T = .36 * 238137.19^{\frac{2}{3}} * 2.80^{\frac{1}{3}} = 1949.96$$

Calculation of h_i

The formula for h_i is the $h_i = \frac{Nu * k}{Tank Diameter} = \frac{1949.96 * 0.645}{.223} = 5640.02$

Calculation of U

The overall heat transfer coefficient is calculated using the values that were calculated earlier:

$$U_i = \frac{Q_{13}}{(t_1 - t_3)A} = \frac{7008}{(\pi * 0.223 * 0.407)(386.15 - 338)} = 510.44 \frac{W}{m^2 K}$$

Where 386.15 is the temperature of the steam (113C)

Vessel area = $\pi \times D \times L$

Data Tabulation/Graphs

Table 1 Experimental Results part A

Impeller Speed RPM	Input hex temp				CW FLOW				HEX FLOW TUBE SIDE			
615	65	C	338	k	0.752	gpm	0.047	kg/s	0.79	gpm	0.050	kg/s
775	65	C	338	k	0.6016	gpm	0.038	kg/s	0.79	gpm	0.050	kg/s
900	65	C	338	k	0.5828	gpm	0.037	kg/s	0.79	gpm	0.050	kg/s

Table 2 Experimental Results part A

Impeller Speed RPM	Ouptut Heat Exchanger				Tank Temp				CW OUT				CW IN			
615	55	C	328	k	65.00	C	338.00	k	46.11	C	319.11	k	21.11	C	294.11	k
775	57	C	330	k	65.83	C	338.83	k	47.22	C	320.22	k	21.11	C	294.11	k
900	57.5	C	330.5	k	65.28	C	338.28	k	50.00	C	323.00	k	21.11	C	294.11	k